

```
#include <ESP8266WiFi.h>

#include <ESP8266WebServer.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>


const char* ssid = "YOUR_WIFI_NAME";

const char* password = "YOUR_WIFI_PASSWORD";


ESP8266WebServer server(80);


LiquidCrystal_I2C lcd(0x27, 16, 2);


#define BUTTON1 D5
#define BUTTON2 D6
#define BUTTON3 D7


int vote1 = 0;

int vote2 = 0;

int vote3 = 0;


void setup()
{
    Serial.begin(115200);


    pinMode(BUTTON1, INPUT_PULLUP);
    pinMode(BUTTON2, INPUT_PULLUP);
    pinMode(BUTTON3, INPUT_PULLUP);
```

```
lcd.init();
```

```
lcd.backlight();
```

```
lcd.setCursor(0, 0);
```

```
lcd.print("IoT Voting");
```

```
lcd.setCursor(0, 1);
```

```
lcd.print("Machine");
```

```
delay(2000);
```

```
lcd.clear();
```

```
lcd.print("Connecting WiFi");
```

```
WiFi.begin(ssid, password);
```

```
while (WiFi.status() != WL_CONNECTED)
```

```
{
```

```
    delay(500);
```

```
    Serial.print(".");
```

```
}
```

```
Serial.println();
```

```
Serial.println("WiFi Connected");
```

```
Serial.print("IP Address: ");
```

```
Serial.println(WiFi.localIP());
```

```
lcd.clear();
```

```
lcd.print("WiFi Connected");
```

```
delay(1000);
```

```
lcd.clear();
```

```
lcd.print(WiFi.localIP());
```

```
delay(3000);
```

```
server.on("/", handleHome);
```

```
server.on("/reset", handleReset);
```

```
server.begin();
```

```
lcd.clear();
```

```
lcd.print("Voting Ready");
```

```
delay(1500);
```

```
showVotes();
```

```
}
```

```
void loop()
```

```
{
```

```
server.handleClient();
```

```
if (digitalRead(BUTTON1) == LOW)
```

```
{
```

```
vote1++;
```

```
lcd.clear();
```

```
lcd.print("Vote Recorded");
```

```
lcd.setCursor(0, 1);
```

```
lcd.print("Candidate 1");
```

```
delay(1500);
```

```
showVotes();
```

```
while (digitalRead(BUTTON1) == LOW);
```

```
}
```

```
if (digitalRead(BUTTON2) == LOW)
```

```
{
```

```
vote2++;
```

```
lcd.clear();
```

```
lcd.print("Vote Recorded");
```

```
lcd.setCursor(0, 1);
```

```
lcd.print("Candidate 2");
```

```
delay(1500);
```

```
showVotes();
```

```
while (digitalRead(BUTTON2) == LOW);
```

```
}
```

```
if (digitalRead(BUTTON3) == LOW)
```

```
{  
    vote3++;  
  
    lcd.clear();  
    lcd.print("Vote Recorded");  
    lcd.setCursor(0, 1);  
    lcd.print("Candidate 3");  
  
    delay(1500);  
  
    showVotes();  
  
    while (digitalRead(BUTTON3) == LOW);  
}  
}  
  
void showVotes()  
{  
    lcd.clear();  
  
    lcd.setCursor(0, 0);  
    lcd.print("C1:");  
    lcd.print(vote1);  
  
    lcd.setCursor(8, 0);  
    lcd.print("C2:");  
    lcd.print(vote2);
```

```
lcd.setCursor(0, 1);  
lcd.print("C3:");  
lcd.print(vote3);  
}
```

```
void handleHome()
```

```
{
```

```
int total = vote1 + vote2 + vote3;
```

```
String webpage = "";
```

```
webpage += "<html>";
```

```
webpage += "<head>";
```

```
webpage += "<title>IoT Voting Machine</title>";
```

```
webpage += "<meta name='viewport' ";
```

```
webpage += "content='width=device-width, initial-scale=1'>";
```

```
webpage += "<style>";
```

```
webpage += "body{";
```

```
webpage += "font-family:Arial;";
```

```
webpage += "text-align:center;";
```

```
webpage += "background:#f2f2f2;";
```

```
webpage += "}";
```

```
webpage += ".box{";
```

```
webpage += "background:white;";
```

```
webpage += "padding:20px;";  
webpage += "margin:20px;";  
webpage += "border-radius:10px;";  
webpage += "}";
```

```
webpage += "button{";  
webpage += "background:red;";  
webpage += "color:white;";  
webpage += "padding:12px;";  
webpage += "border:none;";  
webpage += "border-radius:5px;";  
webpage += "}";
```

```
webpage += "</style>";  
webpage += "</head>";
```

```
webpage += "<body>";
```

```
webpage += "<div class='box'>";
```

```
webpage += "<h1>IoT Voting Machine</h1>";
```

```
webpage += "<h2>Voting Results</h2>";
```

```
webpage += "<p>Candidate 1: ";
```

```
webpage += vote1;
```

```
webpage += " Votes</p>";
```

```
webpage += "<p>Candidate 2: ";
```

```
webpage += vote2;
```

```
webpage += " Votes</p>";
```

```
webpage += "<p>Candidate 3: ";
```

```
webpage += vote3;
```

```
webpage += " Votes</p>";
```

```
webpage += "<h2>Total Votes: ";
```

```
webpage += total;
```

```
webpage += "</h2>";
```

```
webpage += "<br>";
```

```
webpage += "<a href='/reset'>";
```

```
webpage += "<button>RESET VOTES</button>";
```

```
webpage += "</a>";
```

```
webpage += "</div>";
```

```
webpage += "</body>";
```

```
webpage += "</html>";
```

```
server.send(200, "text/html", webpage);
```

```
}
```

```
void handleReset()
```

```
{
```

```
vote1 = 0;
```

```
vote2 = 0;
```

```
vote3 = 0;
```

```
lcd.clear();
```

```
lcd.print("Votes Reset");
```

```
delay(1500);
```

```
showVotes();
```

```
server.sendHeader("Location", "/");
```

```
server.send(303);
```

```
}
```